TEXORA · LECTURE SHEET

Sheet Metal Work and Making Useful Objects

From flat sheet to finished article: materials and tools, layout and development of surfaces, bend allowance, cutting, folding, hemming, seaming and wiring edges — and the making of a tray, funnel and mug as complete worked jobs.

ME-1208 · ENGINEERING WORKSHOP · TOPIC 3 · LEVEL 1 TERM 2

1. Materials and tools

Workshop sheet work uses mild steel (often galvanised or tin-plated), aluminium, copper, brass and stainless sheet, specified by thickness (mm, or gauge — higher gauge number = thinner sheet). The kit: **snips** (straight and curved; aviation snips are colour-coded — red cuts left curves, green right, yellow straight), bench shears and guillotine for straight production cuts; **stakes** (anvil shapes: hatchet, half-moon, funnel, round-head) over which sheet is shaped with **mallets** and wooden or rawhide hammers that do not stretch or mark it; folding bars and a bench folder for crisp bends; scriber, dividers and templates for layout.

2. Layout — development of surfaces

A sheet-metal article is made from its **development**: the 3D surface unrolled flat. Three classical methods (the theory was built in ME-1102): **parallel-line** development for prisms and cylinders (a tray, a pipe — the flat width is the perimeter, πD for a cylinder); **radial-line** for cones and pyramids (all lines radiate from the apex; the pattern is a circular sector whose arc is the base circumference); **triangulation** for transition pieces (square-to-round). Allowances are added at layout: for hems, seams and wired edges (below), and bends consume material by the bend allowance.

3. Bending and the bend allowance

When sheet bends, the outside stretches and the inside compresses; between them lies the **neutral axis**, un-stretched, at roughly (0.3–0.5) × thickness from the inside face (factor K). Flat length must be computed along the neutral axis:

$$\text{Bend allowance } BA = (\theta \text{ in radians}) \times (R + K\,t)$$

WORKED EXAMPLE

A 90° bend in 1.5 mm sheet, inside radius R = 3 mm, K = 0.4: neutral radius = 3 + 0.4 × 1.5 = 3.6 mm; θ = π/2. BA = 1.571 × 3.6 = **5.65 mm**. The flat pattern is the sum of the straight legs (measured to the tangent points) plus 5.65 mm for the bend — ignore it, and every folded article comes out short.

Bend with the grain where possible, keep the inside radius at least equal to the thickness for mild steel (sharper invites cracking on the outside fibres), and springback — the bend opening a few degrees when released — is compensated by slight over-bending.

4. Edges, seams and joints of sheet work

- **Hem:** the edge folded back on itself (allowance ≈ 2 × hem width in layout) — removes the knife edge and stiffens it.
- **Wired edge:** the edge rolled around a wire core: strongest and safest rim (bucket and mug rims), allowance ≈ 2.5 × wire diameter.
- **Grooved (lock) seam:** two hooked edges folded together and hammered flat — the classic longitudinal seam of buckets and pipes, then soldered watertight if needed.
- **Pittsburgh seam, standing seam:** folded joints for ducts and roofing.
- Soldered, riveted or spot-welded laps where folding will not serve (joining processes proper are Topic 4).

5. Three complete jobs

ARTICLE	DEVELOPMENT & SEQUENCE
Open tray 200 × 150 × 40 mm	Flat blank (200 + 2 × 40) × (150 + 2 × 40) = 280 × 230 mm; cut away the four 40 × 40 corner squares; fold four sides up on the folder (check BA if thicknesses warrant); solder or spot-weld the corner seams; hem the rim. Cutting corners first is the whole trick — fold lines are scribed before any cut
Funnel (frustum, top Ø120, throat Ø30, height 90)	Radial-line development: extend the frustum to its apex, two arcs (slant radii L and l), sector angle from the base circumference 2πr = θL. Add seam allowance on one radial edge, hem the top rim, wire the throat if it takes a handle load; form round on the funnel stake, lock-seam and solder
Cylindrical mug Ø80 × 100 mm	Body blank = π × 80 = 251.3 mm long (+ seam allowance) × 100 mm (+ wired-edge allowance at top, base-seam allowance at bottom); base disc Ø80 + flange allowance;

	roll the body on bending rolls/a pipe former, grooved-seam the side, solder the base flange in, wire the top edge, rivet/solder the handle strap

6. Key points

- Developments: parallel-line (prisms/cylinders), radial-line (cones/pyramids), triangulation (transitions) — plus allowances for bends, hems, seams and wiring.
- BA = θ(R + Kt), θ in radians; neutral axis sits K ≈ 0.3–0.5 of the thickness from the inside.
- Hems kill sharp edges; wired edges add a stiff safe rim (≈ 2.5 d allowance); grooved seams join sheet to itself mechanically.
- Inside bend radius ≥ sheet thickness for mild steel; expect springback and over-bend slightly.
- Jobs follow: layout → cut (corners/openings first) → fold → seam → finish edges → join fittings.

7. Practice

Answer in your own words; brief model answers follow.

1. Find the bend allowance for a 90° bend in 2 mm sheet with inside radius 4 mm and K = 0.35.
2. A square tray has a 160 × 160 mm base and 30 mm sides, from one blank with corner cut-outs and no bend-allowance correction. Give the blank size and the corner square size.
3. For the mug of Section 5, why is the body blank longer than the mug's circumference by more than the πD figure, and what two allowances make up the difference?
4. A funnel is developed by the radial-line method. State what the two radii of the pattern are and how the sector angle is fixed.
5. Why must the inside radius of a bend in sheet not be made very sharp, and what defect appears if it is?
6. You need red, green and yellow aviation snips for one job. Assign each to its cut: straight line, clockwise curve, anti-clockwise curve.

Answer key

1. Neutral radius = 4 + 0.35 × 2 = 4.7 mm; θ = π/2 = 1.571 rad. BA = 1.571 × 4.7 = 7.38 mm.
2. Blank = (160 + 2 × 30) × (160 + 2 × 30) = 220 × 220 mm; the four corner squares are 30 × 30 mm each.
3. The πD = 251.3 mm covers only the finished circumference measured on the neutral axis of the finished cylinder; the flat blank additionally needs the grooved-seam allowance along one vertical edge (two hooked folds that interlock) and the wired-edge allowance at the top rim (≈ 2.5 × wire

diameter) — plus the base-seam flange allowance along the bottom edge. Allowances are where flat sheet pays for its joints.

4. The radii are the slant distances from the cone’s apex to the frustum’s two rims (top and throat), obtained by extending the frustum’s side profile to the apex. The sector angle is fixed by rolling the base circumference onto the outer arc: angle (radians) = base circumference / outer slant radius = $2\pi R_{top}/L$.
5. A sharp inside radius forces the outer fibres around a tiny arc: local strain exceeds the sheet’s ductility and the bend cracks (or thins dangerously) on its outer surface. Gentle radii (≥ thickness for mild steel, more for hard or brittle stock) spread the strain over a longer neutral-axis length.
6. Yellow — straight cuts (and gentle curves). Red — anti-clockwise/left curves (the red pair is ground to cut left-hand curves comfortably). Green — clockwise/right curves. The colours are an industry convention so the right snip is grabbed without inspection.
